Site report - *Summer 2015*:

***NorSource* - related work**

Research Group in Digital Linguistics

NTNU, Trondheim

http://www.ntnu.edu/digital-linguistics

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(shorter version presented by Dan Flickinger)

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NTU, Singapore

Activities in the Research Group

are shared between **TypeCraft** and **NorSource;** here we present recent work on NorSource (‘Norwegian HPSG Resource Grammar’).

Main presentation site:

http://typecraft.org/tc2wiki/Norwegian_HPSG_grammar_NorSource

Web demo:

http://regdili.hf.ntnu.no:8081/linguisticAce/parse

NorSource - stages

The grammar since its start in 2002 has gone through the following stages:

- Phase 1, the *Grounding* phase (2001-04),
- Phase 2, the *Semantic Expansion* phase (2005-07),
- Phase 3, the *Cross-Linguistic Coding* phase (2008-10), and
- Phase 4, the *Interoperability* phase (2010-).
- **Phase 1** resided in the building of a basic core grammar around the Matrix skeleton (using the Matrix versions 0.1 – 0.6, as they developed; this included the MRS system). This stage included the accommodation of a 80,000 entries lexicon imported from the previously established resources TROLL and NorKompLex, where a verb valence code and a code for inflectional paradigms constituted major parts. Main publications from this period are: Hellan and Haugereid 2002, Hellan 2003.
- **Phase 2** resided in the development of a fine-grained ontology and computing system of spatial and temporal relations, amenable to grammatical systems across languages and typologies, and a detailed semantics of comparative constructions. The grammar was also used as a part of a small Norwegian-Japanese MT system. In this period, the inflectional system was thoroughly revised. Main publications : Hellan and Beermann (2004), Bermann et al. (2004), Beermann and Hellan (2005), Hellan and Beermann (2012).

A tdl-file with the semantics of spatial and temporal relations for prepositions and adverbs can be downloaded from http://typecraft.org/tc2wiki/Norwegian_HPSG_grammar_NorSource ; this file is intended to fit into any current Matrix grammar (recently integrated into BURGER, the Bulgarian grammar), and includes sample lexical entries for the relevant preposition types.

NorSource - stages

Phase 3 was devoted to a thorough revision of the valence code, to accommodate a cross-linguistically defined classification system of valence and construction types. Main publications : Hellan (2008), Hellan and Dakubu (2010), Dakubu and Hellan forthcoming. Opens also for *Grammar Induction*, see slide 15.

Phase 4 has resided in the development of applications:

- A 'Grammar Sparrer', as described in Hellan et al. 2013, accessed at A Norwegian Grammar Sparrer. This is a construct along the lines of Bender et al. 2004, and Suppes et al. 2014, falling within the overall initiatives described in Heift and Schultze 2007, where specific types of grammatical mistakes are accommodated by 'mal-rules' in an extended 'mal'-version of the grammar, and parses involving such mal-phenomena are reported to the user as tutoring instructions. This system has been running as a webdemo since 2011. See slides.
- A Multilingual Valence repository, called ***MultiVal***, based on NorSource and three further LKB grammars: The Spanish Resource Grammar, the Bulgarian grammar BURGER, and a grammar of Ga. See slides below.
- An initial version of a POS-tagger of Norwegian, reflecting the lexical inventory of the grammar, which amounts to appx. 85000 lexical entries, and a large number of proper names of various categories. The tagger currently offers all available POS-alternatives for a given word. See web access at http://regdili.hf.ntnu.no:8081/webtagger/tagger. See slide 22.

This POS-tagger is part of a **Phase 5**, which aims at enabling the grammar to parse larger texts and longer sentences, which so far have not been prioritized areas. A series of slides exemplify recent initiatives in this direction, conducted by Tore Bruland (no publications yet) – see below.

Application 1. Constructing an e-learning tool from an LKB grammar

The ***Norwegian Online Grammar Sparrer*** is an online language training tool developed at NTNU, with a direct access point at

http://regdili.hf.ntnu.no:8081/studentAce/parse

and a wiki access point at

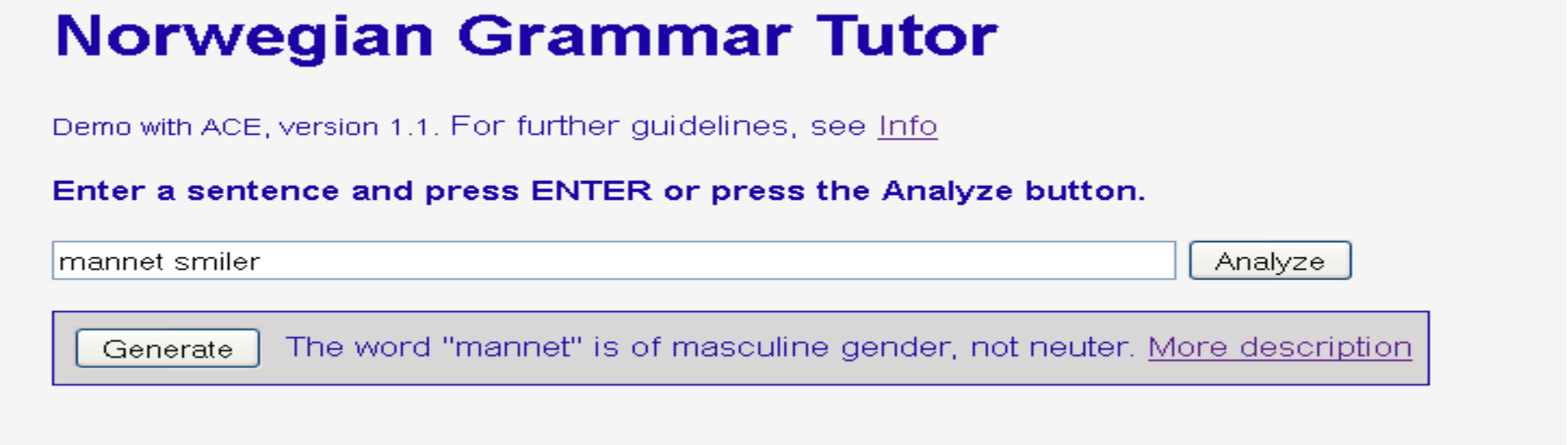
http://typecraft.org/tc2wiki/A_Norwegian_Grammar_Sparrer

An introduction to its 'mal-grammar'-based design is given in Hellan et al. 2013.

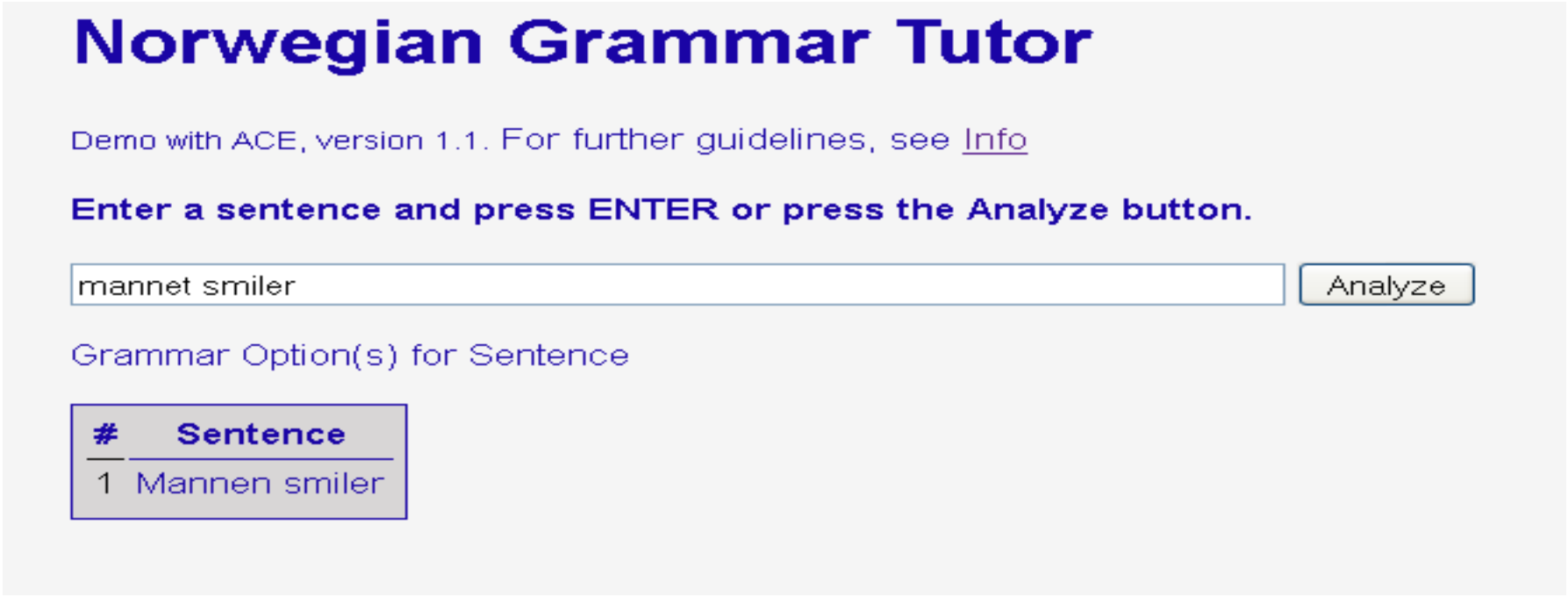
Its basics, as developed in 2011-2013, are indicated on the following two slides:

The procedure - 1

- If you enter an ungrammatical sentence …
- you receive an error message (see lowest line underneath).
- If the MRS constructed for the sentence by the ’mal-grammar’ is wellformed, a button for ’Generate’ appears (see below), by which a ’correct’ version of the sentence can be evoked.



The procedure – 2: ’Generate’ to see acceptable option



During the last year

The interface now accepts ***batches of up to 10 sentences***, each with max. 10 words

Responses are given not only in English, but also in ***Polish*** (thanks to Kasia Klessa, Univ. of Poznan) More 'response' languages can be added.

The design with ***freely chosen inputs*** is maintained. (This requires a large grammar and lexicon – 85 000 entries.)

The number of actions (processing a batch of sentences, or doing a generation) has been around 200 per day during 2015.

A corpus of input strings is being accumulated. (But we keep no track of users.)

The system now sits in a virtual server at the faculty of Humanities, NTNU.

The system functions on a Creative Common license basis.

Example of batch of 10 sentences with responses in English:

Norwegian Grammar Tutor

Demo with ACE, version 1.11. For further guidelines, see Info

Enter up to 10 sentences with up to 10 words each and press the Analyze button.

Feedback Language English

```
Du liker jeg.
Jeg prøver og komme.
Ola å Per kommer.
Jeg skammer seg.
Imorgen jeg kommer.
Kommer jeg snart.
Jeg like fisken.
Jeg prøvde å gikk.
Husen er gult.
En gult bil stod her.
```

Analyze

generate	The word "jeg" is marked with the wrong case, try using "meg" instead. More description
generate	The word "og" is not the infinitival marker, try using "å" instead. More description
3	**Ungrammatical in Norwegian**
generate	The reflexive pronoun "seg" does not match the number and gender of the word it refers back to. Try using "meg" instead. More description
generate	The sentence lacks subject-verb inversion. More description
generate	The sentence contains an incorrect subject-verb inversion. More description
generate	The word "like" has the wrong tense.. More description
8	**Ungrammatical in Norwegian**
generate	The word "husen" is of neuter gender, not masculine. More description
generate	The adjective "gult" is conjugated as neuter gender, but modifies a masculine or feminine noun. More description

Example of batch of 10 sentences with responses in Polish:

Norwegian Grammar Tutor

Demo with ACE, version 1.11. For further guidelines, see Info

Enter up to 10 sentences with up to 10 words each and press the Analyze button.

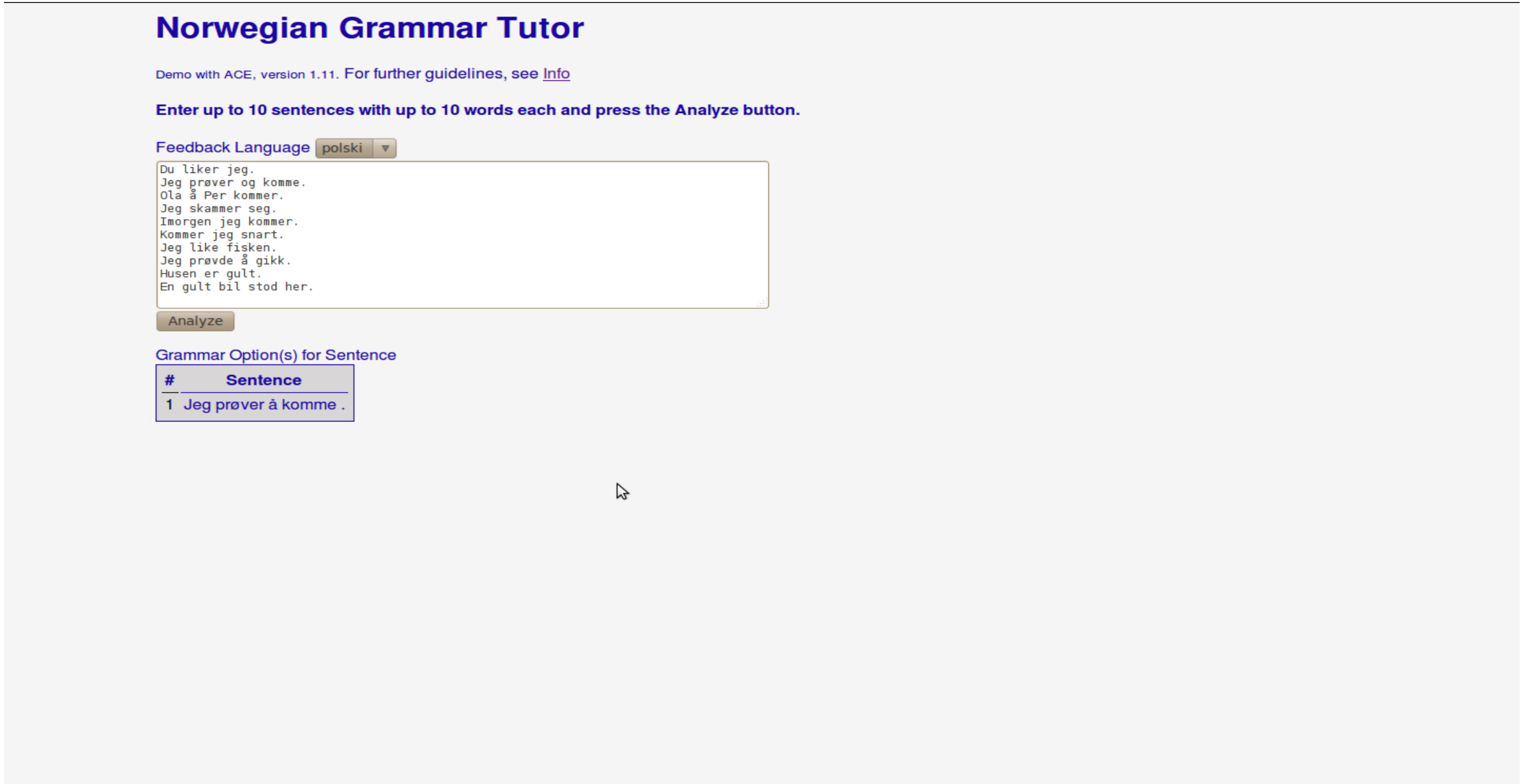
Feedback Language polski

```
Du liker jeg.
Jeg prøver og komme.
Ola å Per kommer.
Jeg skammer seg.
Imorgen jeg kommer.
Kommer jeg snart.
Jeg like fisken.
Jeg prøvde å gikk.
Husen er gult.
En gult bil stod her.
```

Analyze

generate	Wyraz "jeg" jest w nieprawidłowym przypadku, spróbuj użyć "meg". More description
generate	Wyraz "og" nie jest markerem bezokolicznikowym, spróbuj użyć "å". More description
3	**Ungrammatical in Norwegian**
generate	Zaimek zwrotny "seg" nie zgadza się z liczbą i rodzajem wyrazu, do któego się odnosi. Spróbuj użyć "meg". More description
generate	W tym zdaniu brakuje inwersji podmiot-czasownik. More description
generate	To zdanie zawiera nieprawidłową inwersję (odwróconą kolejność) podmiot-czasownik. More description
generate	Wyraz "like" jest w nieprawidłowym czasie. More description
8	**Ungrammatical in Norwegian**
generate	Wyraz "husen" jest rodzaju nijakiego, a nie męskiego. More description
generate	Przymiotnik "gult" odmienia się jak w rodzaju nijakim, ale określa rzeczownik męski lub żeński. More description

Using the button '*Generate*' to see acceptable option



Application 2. ***MultiVal*** – the Multilingual Valency database

The system by now hosts 4 languages, with altogether 40 000 verb entries, with valency frames classified in a uniform system. The languages hosted are:

Bulgarian *(lexicon import from BURGER, the Bulgarian Matrix grammar)*

Ga *(lexicon import from GaGram, the Ga Matrix grammar, whose lexicon is in turn imported from a ToolBox lexicon of Ga, created by M.E.Kropp Dakubu)*

Norwegian *(lexicon import from NorSource , the Norwegian Matrix grammar)*

Spanish *(lexicon import from SRC , the Spanish Matrix grammar)*

For documentation of the system per February 2014 (before Bulgarian got added), see Hellan et al., LREC 2014.

The following slide shows search results for ‘intransitive’, for verb starting with “s”.

The next slide in turn shows information as it looks for a given verb, and shows two features of interoperability with other applications – *TypeCraft* and *ImagAct*.

For the button SHOW:

Automatic import of urls for glossed examples from ***TypeCraft*** has been defined, and links to ***ImagAct*** scene videos are being added – here for *Marit tar seg på kinnet:*

Language	Norwegian Bokmål
Verb id	ta_tr-detachposs-refl
Syntactic Arguments	NP+NPrefl+PP
FCT	transReflxWithOblique
SIT	ternaryPossessorDetachment
Aspect	
Verb type	v-trObl-obRefl_oblPRTOFob
Example of type	Ola klør seg på ryggen
Orthography	ta
English gloss	[take] – only through TypeCraft link
Example	[Marit tar seg på kinnet] – only through TypeCraft link
Free translation	[Mary touches her cheek] – only through TypeCraft link
TypeCraft URL	http://typecraft.org/tc2/ntceditor.html#2790,45468
ImagAct URL	http://www.imagact.it/imagact/sceneMetadata.seam?sceneId=54&cid=9995

Version 1.4 (for further guidelines, see Info)

Languages:

☑ Norwegian ☑ Ga ☑ Spanish ☑ Bulgarian

Search fields:

Verb lexeme: s

Syntactic Arguments:

Function: intransitive

Situation:

Aspect:

Type:

Search | Count | Clear | Download

Ga	show	sa	NP
Ga	show	sa	NP
Norwegian Bokmål	show	sabbe_intr-dir	NP
Norwegian Bokmål	show	sabbe_iv	NP
Norwegian Bokmål	show	safe_iv	NP
Norwegian Bokmål	show	safte_iv	NP
Norwegian Bokmål	show	sage_iv	NP
Norwegian Bokmål	show	sake_iv	NP
Norwegian Bokmål	show	sakke_iv	NP
Norwegian Bokmål	show	sakse_iv	NP
Norwegian Bokmål	show	saktne_iv	NP
Spanish	show	salivar_v	NP
Norwegian Bokmål	show	saluttere_iv	NP
Bulgarian	show	salyutiram_v1	+
Norwegian Bokmål	show	samarbeide_iv	NP
Norwegian Bokmål	show	sameksistere_iv	NP
Bulgarian	show	samoobladavam_v1	+

For the button SHOW:

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Aspect	
Verb type	v-trObl-obRefl_oblPRTOFob
Example of type	Ola klør seg på ryggen
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English gloss	take
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Grammar Induction

Our package for Grammar Induction from IGT with valency annotation (see Hellan (2010), Bruland (2011), and Hellan and Beermann (2011, 2014))

is now downloadable from:

http://regdili.hf.ntnu.no:8081/typegramusers/menu

and also via – where an introduction is given:

http://typecraft.org/tc2wiki/TypeGram

‘Phase 5’:

Preprocessing Text

Increasing performance and robustness

With components based on the NorSource grammar

Motivation

- The long term goal is to create a Natural Language Processing (NLP) pipeline for Norwegian, where the input is correct written Bokmål or Nynorsk
- The short term goal is to process documents and provide a meaning representation from Syntax or Lexical Semantics
- First, we focus on preprocessing text before the parsing process (both shallow and deep). Then, we can integrate DELPH-IN tools into the pipeline

Get the Text

We read RSS feeds from NRK (Norwegian Broadcast Company) and find the text in the HTML tags:

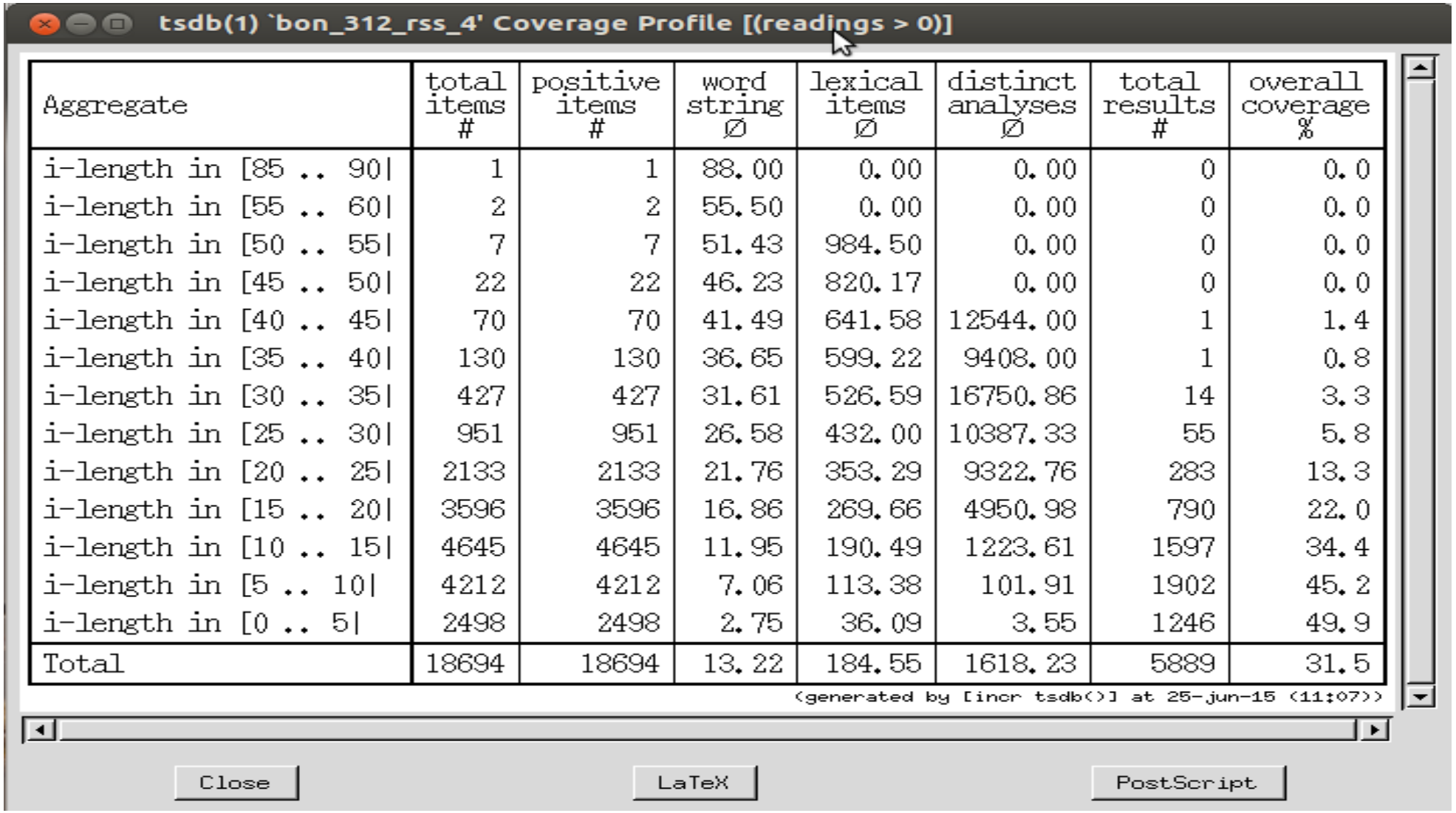
```
h2: Sogn og Fjordane
h1: - Selgarar prøver å lure folk med kreftfrykt
p: Dei siste dagane har eit firma ringt rundt i Høyanger for å selje ein
   radontest. Testen er langt over vanleg pris, og seljarane lyg om at
   dei samarbeider med kommunen.
p: MANGE TELEFONAR: Mange folk i Høyanger har fått telefonar frå
   eit firma som vil selje radontestar dei siste dagane.
```

Then, we find the sentence borders

A small parsing experiment

- We have collected 1128 HTML documents
- We used Ordbank Bokmål and Ordbank Nynorsk to classify a document as Bokmål or Nynorsk
- 95 of the documents were Nynorsk, and were omitted
- The remaining 1033 documents contain 18694 sentences
- We have created some extra words from these sentences and put them in an extra lexicon
- The sentences were parsed with Pet from The LOGON software

Coverage from [incr tsdb()]



Aggregate	total items #	positive items #	word string Ø	lexical items Ø	distinct analyses Ø	total results #	overall coverage %
i-length in [85 .. 90]	1	1	88.00	0.00	0.00	0	0.0
i-length in [55 .. 60]	2	2	55.50	0.00	0.00	0	0.0
i-length in [50 .. 55]	7	7	51.43	984.50	0.00	0	0.0
i-length in [45 .. 50]	22	22	46.23	820.17	0.00	0	0.0
i-length in [40 .. 45]	70	70	41.49	641.58	12544.00	1	1.4
i-length in [35 .. 40]	130	130	36.65	599.22	9408.00	1	0.8
i-length in [30 .. 35]	427	427	31.61	526.59	16750.86	14	3.3
i-length in [25 .. 30]	951	951	26.58	432.00	10387.33	55	5.8
i-length in [20 .. 25]	2133	2133	21.76	353.29	9322.76	283	13.3
i-length in [15 .. 20]	3596	3596	16.86	269.66	4950.98	790	22.0
i-length in [10 .. 15]	4645	4645	11.95	190.49	1223.61	1597	34.4
i-length in [5 .. 10]	4212	4212	7.06	113.38	101.91	1902	45.2
i-length in [0 .. 5]	2498	2498	2.75	36.09	3.55	1246	49.9
Total	18694	18694	13.22	184.55	1618.23	5889	31.5

(generated by [incr tsdb()] at 25-jun-15 (11:07))

Challenge

- The texts have more names, terms and compound words than the lexicon, and if we enter them into the lexicon, our grammar will grow and use more resources.
- We put the words in a database and we find them in a preprocessing step.
- We need to find unknown words and give them a definition.
- We use a tagger and a compound word splitter.

Creating a Tagger from NorSource

- We unload the type hierarchy from ACE
- We read the inflection rules
- We assign a Part of Speech at some nodes in the hierarchy
- We use the lexicon type to find a PoS
- early version: http://regdili.hf.ntnu.no:8081/webtagger/tagger

We create a table with a parent and all its children. We do not store how far down the child occurs.

The type nominal_inflection has key=6 and all its children are shown in the blue list.

Complex example.
- the rule = final-full
- the lexicon entry = infl_nshort

- a common child type

```
ij> select a.parent_k, a.child_k, b.infl_na
from sa.infl_hierarki a,
     sa.inflection b
where parent_k = 6 and b.infl_k = a.child_k;
PAREN&|CHILD&|INFL_NA
--------------------------------
6     |5     |adjective_inflection
6     |6     |nominal_inflection
6     |7     |final-full-nlong
6     |8     |final-full-n
6     |9     |final-full
6     |10    |final-full-long-or-&
6     |11    |infl_m_f
6     |12    |noun_inflection
6     |13    |final-full-nshort
6     |14    |g(72)
6     |15    |infl_mshort
6     |16    |infl_m
6     |17    |infl_fshort
6     |18    |infl_f
6     |19    |infl_m_fshort
6     |20    |final-full-mm-nlong
6     |21    |final-full-mm
6     |22    |final-full-mm-nshort
6     |23    |g(77)
6     |24    |g(71)
6     |25    |g(75)
6     |26    |g(73)
6     |27    |g(74)
6     |28    |light-e
6     |29    |light-e-mf-sup
6     |30    |light-e-short
6     |31    |infl_m_nshort
6     |32    |infl_nshort
6     |33    |light-e-nshort
6     |34    |infl_n
6     |35    |light-e-nlong
6     |36    |g(76)
6     |37    |infl_nlong
6     |38    |light-e-ic
6     |39    |light-e-sup

35 rows selected
```

```
 6  2     adjective_inflection
 7  2     noun_inflection
 8  3       infl_m_f
 9  4         final-full-long-or-short
10  5           final-full
11  6             final-full-n
12  7               final-full-nlong
13  7               final-full-nshort
14  5           g(72)
15  6             final-full-nshort
16  6             infl_m
17  7               infl_mshort
18  ...
19  4         final-full-mm
20  5           final-full-mm-nlong
21  5           final-full-mm-nshort
22  5           g(77)
23  6             final-full-mm-nlong
24  6             final-full-mm-nshort
25  ...
26  4         light-e-mf-sup
27  5           light-e
28  5           light-e-short
29  3       infl_m_nshort
30  4         g(71)
31  5           g(72)
32  6             final-full-nshort
33  6             infl_m
34  7               infl_mshort
35  5           g(75)
36  6             final-full-mm-nshort
37  6             final-full-nshort
38  4         infl_m
39  5           infl_mshort
40  4         infl_nshort
41  5           final-full-mm-nshort
42  5           final-full-nshort
43  5           g(75)
```

```
sg_def_n_final-full_irule :=
  %suffix (* et)(ium iet)
  neutsg-def-noun-word &
   [ ARGS < sg-neut-def-noun-lexeme &
          [ SYNSEM.LOCAL.CAT.HEAD [DEF +,
                                   DEFINITE +],
            INFLECTION  final-full & [NUMB sing, DEFINITENESS def, GEND n] ] > ].
```

We read the inflection rule files and store them in a table. We assign a PoS to the rule.

```
select * from sa.rule where rule_k = 14;
RULE_K|PATTE&|REGU&|INFL_K|POS_K |STEM_AFFIX  |AFFIX
------------------------------------------------------
14    |1     |true |3     |11    |*           |et
14    |2     |true |3     |11    |ium         |iet

2 rows selected
```

```
fyr_n_neut := neut-re-noun-lxm &
 [ INFLECTION infl_nshort,
   STEM < "fyr" >,
   SYNSEM [ LKEYS.KEYREL.PRED "_fyr_n_rel",
            LOCAL.CAT.HEAD.KEYS.KEY anonym-k ] ].
```

We read the lexicon files and store them in a table. We assign a PoS to some general Types in the hierarchy and find a PoS for each lexicon entry.

```
LEX_K      |STEM                |TYPE                |PARENT              |INFL_NA             |POS
-------------------------------------------------------------------------------------------------
28460      |fyr                 |fyr_n_neut          |neut-re-noun-lxm    |infl_nshort         |N
```

Multi-Language Tagger

- The system contains the following taggers: NorSource, Ordbank Bokmål (database from the University of Oslo), Ordbank Nynorsk (database from the University of Oslo), Statens Kartverk (All map references in Norway), WordNet and names.
- The taggers can be use in sequence or in parallel.
- We use two different levels of tags: the usual Part-Of-Speech, and a deeper tag. The name “Washington” can be a Na or national capital, city, state, federal government, President of the United States, or family name (list from WordNet).

Sentence: Thomas Myhre liker 'Kingdom of Heaven' og '12.50 er ikkje nok'

Different taggers
n – Norsource
c – Statens Kartverk
a – Names
w – WordNet
s – Ordbank Nynorsk

Deeper definition

Normal PoS

Tokens

t	token	type	beg	end
b	thoma	ALPHA	0	5
b	s	ALPHA	5	6
b		WHITE	6	7
b	myhre	ALPHA	7	12
b		WHITE	12	13
b	liker	ALPHA	13	18
b		WHITE	18	19
b	'	PUNCT	19	20
b	kingdom	ALPHA	20	27
b		WHITE	27	28
b	of	ALPHA	28	30
b		WHITE	30	31
b	heaven	ALPHA	31	37
b	'	PUNCT	37	38
b		WHITE	38	39
b	og	ALPHA	39	41
b		WHITE	41	42
b	'	PUNCT	42	43
b	12	NUM	43	45
b	.	PUNCT	45	46
b	50	NUM	46	48
b		WHITE	48	49
b	er	ALPHA	49	51
b		WHITE	51	52
b	ikkje	ALPHA	52	57
b		WHITE	57	58
b	nok	ALPHA	58	61
b	'	PUNCT	61	62

Arcs

t	token	type	beg	end
a	thomas	[first_name_n1, surname_n1]	0	6
n	thomas	[Np]	0	6
n	thomas myhre	[Np]	0	12
n	s	[PNposs]	5	6
c	myhre	[Np]	7	12
a	myhre	[surname_n1]	7	12
n	myhre	[Np]	7	12
n	liker	[N,V]	13	18
n	'	[QUOT]	19	20
w	kingdom	[n]	20	27
w	heaven	[n]	31	37
n	'	[QUOT]	37	38
n	og	[CONJ]	39	41
n	'	[QUOT]	42	43
n	12	[CARD]	43	45
n	.	[EndPctuatn]	45	46
n	50	[CARD]	46	48
n	er	[V]	49	51
s	ikkje	[adv]	52	57
n	nok	[ADVs,DEG,QUANT]	58	61
n	'	[QUOT]	61	62

Unknown Words

Token	beg,end	
of	28	30

- The Compound Word Splitter finds possible solutions for compound words based on the NorSource tagger
- We use the word “kloakkmysteriet” “the mystery in the sewer”
- The column “C1” holds the number of characters from the string; the column “C2” holds a number that favors small amounts of words.
- We can also rank them according to well known patterns

C1	C2	Elements
15	98	kloakk/N, mysteriet/N
15	97	klo/N, akk/V, mysteriet/N
15	96	klo/N, ak/V, km/N, ysteriet/N
15	96	kloakk/N, myste/V, ri/V, et/ART
15	95	klo/N, ak/V, km/N, ysteri/N, et/ART
15	95	klo/N, akk/V, myste/V, ri/V, et/ART
15	95	kloakk/N, my/N, ste/N/V, ri/V, et/ART
15	95	kloakk/N, my/N, ster/V, i/P, et/ART
15	95	kloakk/N, mys/V, te/N/V, ri/V, et/ART
15	95	kloakk/N, mys/V, ter/V, i/P, et/ART
15	95	kloakk/N, myst/V, e/PNposs, ri/V, et/ART
15	95	kloakk/N, myst/V, er/V, i/P, et/ART
15	94	klo/N, ak/V, km/N, yste/V, ri/V, et/ART
15	94	klo/N, ak/V, km/N, yster/V, i/P, et/ART
15	94	klo/N, akk/V, my/N, ste/N/V, ri/V, et/ART
15	94	klo/N, akk/V, my/N, ster/V, i/P, et/ART
15	94	klo/N, akk/V, mys/V, te/N/V, ri/V, et/ART
15	94	klo/N, akk/V, mys/V, ter/V, i/P, et/ART
15	94	klo/N, akk/V, myst/V, e/PNposs, ri/V, et/ART
15	94	klo/N, akk/V, myst/V, er/V, i/P, et/ART
15	94	kloakk/N, my/N, s/PNposs, te/N/V, ri/V, et/ART
15	94	kloakk/N, my/N, s/PNposs, ter/V, i/P, et/ART
15	93	klo/N, ak/V, km/N, yst/V, e/PNposs, ri/V, et/ART
15	93	klo/N, ak/V, km/N, yst/V, er/V, i/P, et/ART
15	93	klo/N, akk/V, my/N, s/PNposs, te/N/V, ri/V, et/ART
15	93	klo/N, akk/V, my/N, s/PNposs, ter/V, i/P, et/ART

Create compound words with The Compound Word Splitter

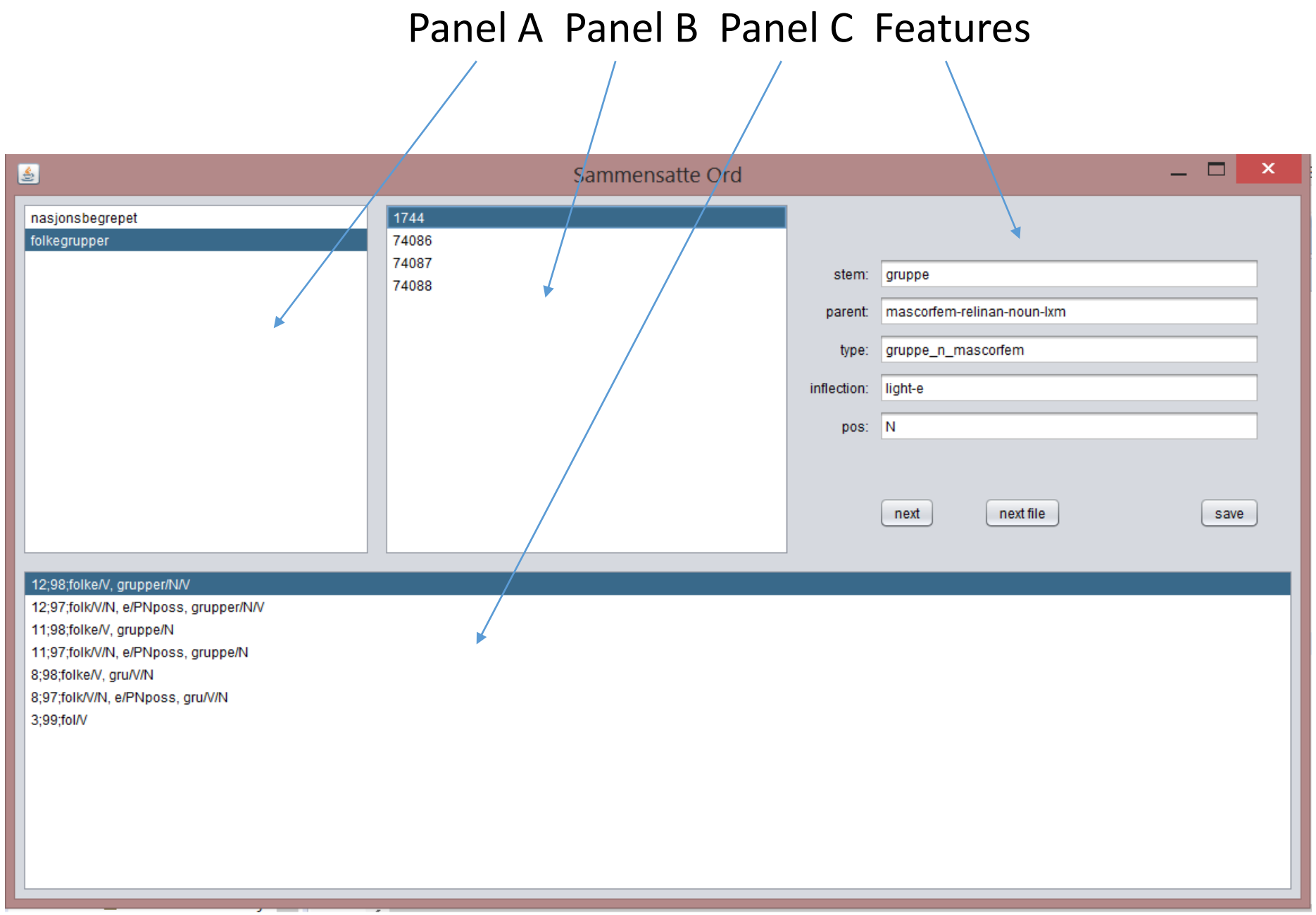
Step 1. An unknown word is found in a sentence from a document and placed in panel A

Step 2. Choose a word and the Compound Splitter result is presented in panel C

Step 3. Choose a compound word and the system puts the NorSource Ids in panel B for the last word

Step 4. Choose an Id and the system puts the data in Features

Step 5. Change the data and press the Save button

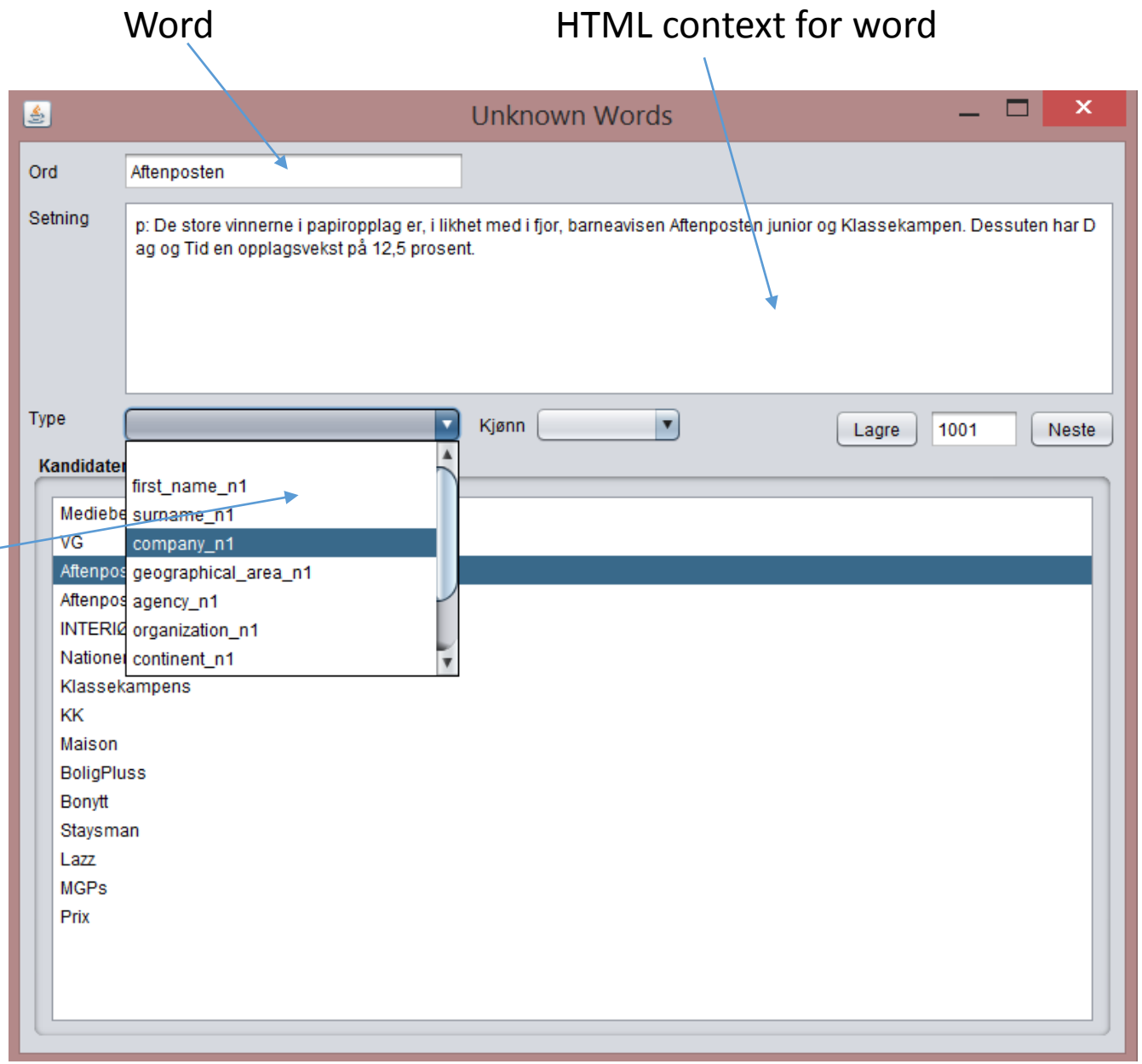


Define Unknown Word with WordNet types

Step 1. Find unknown words in a HTML tag from a document that starts with a capital letter

Step 2. Choose a word and find a type from WordNet and a possible gender

Step 3. Save the word with its new data



Create CoNLL format from NorSource

- We use the Syntax tree and the MRS
- We use the type hierarchy to find if a rule has “head-final” as a parent or if it has “head-initial” as a parent. We then know if the head is in arg1 or arg2.
- We find the Part-Of-Speech from the lexicon table we created for the NorSource tagger
- We connect the tree and the MRS. The tree has token numbers and the MRS has string numbers. We can sort the string numbers and connect them to the token numbers. The lexicon contains the predicate used in the EP.

The sentence: turen går over fjellet

```
Features
(e2,E.TENSE,PRES)
(e2,E.MOOD,INDICATIVE)
(x4,PNG.NG.GEN,M)
(x4,PNG.NG.NUM,SING)
(x4,PNG.PERS,THIRDPERS)
(x11,PNG.NG.NUM,SING)
(x11,PNG.NG.GEN,N)
(x11,PNG.PERS,THIRDPERS)
```

LEXICON

HEAD

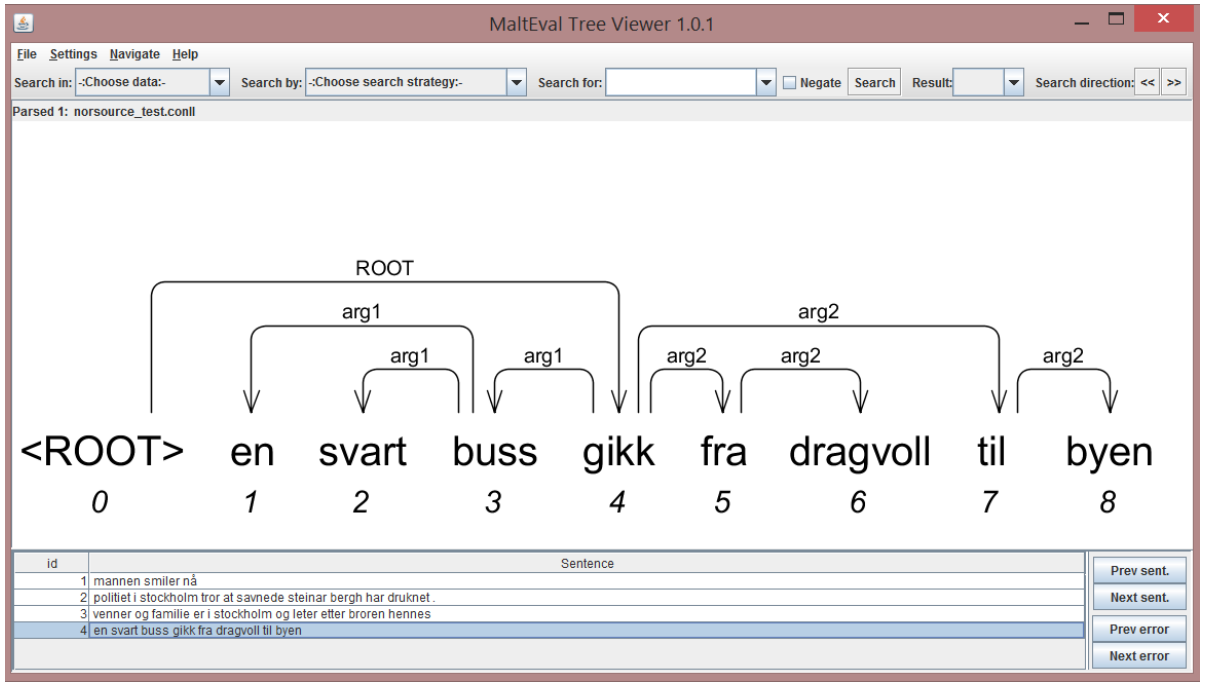
```
Mrs
ltop=h1, index=e2
h3:_tur-1_n_rel([ARG0:x4]), (0,5)
h5:_def_q_rel([ARG0:x4, RSTR:h6, BODY:h7]), (0,5)
h8:_gå_v-intr_rel([ARG0:e2, ARG1:x4]), (6,9)
h8:_over_p-loc_rel([ARG0:u10, ARG1:e2, ARG2:x11]), (10,14)
h12:_fjell_n_rel([ARG0:x11]), (15,22)
h13:_def_q_rel([ARG0:x11, RSTR:h14, BODY:h15]), (15,22)
< qeq(h6, h3), qeq(h14, h12) >
```

```
head-subject-rule (0,4)
  sg_def_m_final-full_irule (0,1)
    sg-masc-def-noun-lxm-lrule (0,1)
      tur_mascdir (0,1)
        TUR
  atelic-pp-mod-vp-index-sit-rule (1,4)
    pres-infl_rule (1,2)
      gå_intrdir (1,2)
        GÅ
    head-prep-comp-rule (2,4)
      over_reg_p (2,3)
        OVER
      sg_def_n_final-full_irule (3,4)
        sg-neut-def-noun-lxm-lrule (3,4)
          fjell (3,4)
            FJELL
```

ID	FORM	LEMMA	CPOSTAG	POSTAG	FEATS	HEAD	DEPREL
1	TUR	_	N	_	M|SING|THIRDPERS	2	arg1
2	GÅ	_	V	_	PRES|INDICATIVE	0	ROOT
3	OVER	_	P	_	_	2	arg2
4	FJELL	_	N	_	SING|N|THIRDPERS	3	arg2

Malt Parser Experiment

- Generated a test set from a parsed Test Suite
- The ‘Features’ (see previous slide) was not used
- We trained and parsed with Arc-Eager, Nivre with allow_root=true, allow_reduce=false and enforce_tree=false
- A few sentences that are not in the test set are shown in figure



Most recent references

Hellan, L., Tore Bruland, Elias Aamot, Mads H. Sandøy (2013): A Grammar Sparrer for Norwegian. *NoDaLiDa 2013*. *NEALT Proceedings Series, volume 16.*

Hellan, L., D. Beermann, T. Bruland, M.E.K. Dakubu, and M. Marimon (2014) MultiVal: Towards a multilingual valence lexicon. *LREC 2014*.

Hellan, L. and D. Beermann (2011, 2014). Inducing grammars from IGT. Z. Vetulani and J. Mariani (eds.) *Human Language Technologies as a Challenge for Computer Science and Linguistics*. Springer.